

Joseph (Joe) D. Osborn

osbornjd91@gmail.com

(859) 433-8738

<https://jdosbo.github.io>

github.com/osbornjd

Research Interests

High-energy experimental nuclear physics, software and computing in high-energy physics.

Education

University of Michigan, Ph.D. in Physics, 2018.

University of Michigan, M.S. in Physics, 2015.

University of Kentucky, B.S. in Physics, B.S. in Mathematics, 2013.

Software and Computational Fluency

Languages: C/C++, Java, Python, L^AT_EX, Bash

Tools: git, Jenkins, Linux, UML, Doxygen, Docker, Singularity, Serverless/FaaS, pybind

HEP Packages: ROOT, ACTS, GEANT, Fun4All, RooUnfold

Ph.D. Thesis

Nonperturbative factorization breaking and color entanglement effects in dihadron and direct photon-hadron angular correlations in $p+p$ and $p+A$ collisions. J. D. Osborn, University of Michigan, May 29, 2018. [hep-ex/1806.07763](https://arxiv.org/abs/hep-ex/1806.07763).

Research Positions

Associate Research Scientist, Oak Ridge National Laboratory

06/2021-present

Computing and Computational Sciences Directorate

Postdoctoral Research Associate, Oak Ridge National Laboratory

07/2019-05/2021

Computing and Computational Sciences Directorate

Visiting Scholar, University of Michigan

06/2019-06/2020

Department of Physics

Postdoctoral Research Fellow, University of Michigan

06/2018-06/2019

Department of Physics

Graduate Research Assistant, University of Michigan

07/2013-06/2018

Thesis advisor: Christine Aidala

Department of Physics

Undergraduate Research Assistant, University of Kentucky

05/2012-05/2013

Thesis advisor: Renee Fatemi

Department of Physics

Undergraduate Research Assistant, University of Kentucky

Summer 2011

Department of Physics

High School Senior Research Assistant, University of Kentucky

01/2008-05/2009

Thesis advisor: James McDonough

Department of Mechanical Engineering

Research Funding

1. Oak Ridge National Laboratory subcontract through Brookhaven National Laboratory. *Software Development for sPHENIX*. 03/2021-03/2022. PI, \$276k.
2. Oak Ridge National Laboratory subcontract through Brookhaven National Laboratory. *Software Development for sPHENIX*. 03/2020-03/2021. PI, \$132k.

Peer Reviewed Publications with Significant Contribution

Full publication list at end of CV, or search ‘find a J. D. Osborn’ on inspirehep.net

1. X. Ai et al. “A Common Tracking Software Project.” arXiv:2106.13593, submitted to Computing and Software for Big Science.
2. R. Abdul Khalek et al. “Science Requirements and Detector Concepts for the Electron-Ion Collider: EIC Yellow Report.” arXiv: 2103.05419.
3. **J.D. Osborn** et al. “Implementation of ACTS into sPHENIX track reconstruction.” Computing and Software for Big Science 5, 23 (2021).
4. U. A. Acharya et al. (PHENIX Collaboration), “Probing gluon spin-momentum correlations in transversely polarized protons through midrapidity isolated direct photons in $p^\uparrow + p$ collisions at $\sqrt{s} = 200$ GeV. Phys. Rev. Lett. 127, 162001 (2021).
5. U. A. Acharya et al. (PHENIX Collaboration), “Transverse single-spin asymmetries of midrapidity π^0 and η mesons in polarized $p+p$ collisions at $\sqrt{s} = 200$ GeV.” Phys. Rev. D 103, 052009 (2021).
6. C. A. Aidala et al. “Design and beam test results for the 2D projective sPHENIX electromagnetic calorimeter prototype.” IEEE Transactions on Nuclear Science, vol. 68, no. 2, pp. 173-181, Feb. 2021.

7. J. D. Osborn for the sPHENIX Collaboration, “Requirements, status, and plans for track reconstruction at the sPHENIX experiment.” arXiv:2007.00771, peer reviewed Connecting the Dots 2020 Workshop proceedings.
8. U. Acharya et al. (PHENIX Collaboration), “Measurement of jet-medium interactions via direct photon-hadron correlations in Au+Au and d +Au collisions at $\sqrt{s_{NN}} = 200$ GeV.” Phys. Rev. C 102, 054910 (2020).
9. J. D. Osborn for the LHCb Collaboration, “Jet hadronization at LHCb.” PoS High- p_T 2019 (2020) 013.
10. I. Helenius, J. Lajoie, **J. D. Osborn**, P. Paakkinen, H. Paukkunen, “Nuclear gluons at RHIC in a multi-observable approach.” Phys. Rev. D 100, 014004 (2019).
11. A. Aaij et al. (LHCb Collaboration), “Measurement of charged-hadron production in Z -tagged jets in proton-proton collisions at $\sqrt{s} = 8$ TeV.” Phys. Rev. Lett. 123, 232001 (2019).
12. C. Aidala et al. (PHENIX Collaboration), “Nonperturbative transverse momentum broadening in dihadron angular correlations in $\sqrt{s_{NN}} = 200$ GeV proton-nucleus collisions.” Phys. Rev. C 99, 044912 (2019).
13. J. D. Osborn for the PHENIX Collaboration, “PHENIX results on jet modification with π^0 - and photon-triggered two particle correlations in $p+p$, $p(d)$ +Au, and Au+Au collisions.” Nuclear Physics A 982, 591-594 (2019).
14. C. Aidala et al. (PHENIX Collaboration), “Nonperturbative transverse momentum dependent effects in dihadron and direct photon-hadron angular correlations in $p+p$ collisions at $\sqrt{s} = 200$ GeV.” Phys. Rev. D 98, 072004 (2018).
15. C. Aidala et al. (PHENIX Collaboration), “Single-spin asymmetry of J/ψ production in $p+p$, $p+Al$, and $p+Au$ collisions with transversely polarized proton beams at $\sqrt{s_{NN}} = 200$ GeV.” Phys. Rev. D 98, 012006 (2018).
16. J. D. Osborn for the PHENIX Collaboration, “Study of cold and hot nuclear matter effects on jets with direct photon triggered correlations from PHENIX.” Nuclear Physics A 967, 476-479 (2017).
17. A. Adare et al. (PHENIX Collaboration), “Nonperturbative-transverse-momentum effects and evolution in dihadron and direct photon-hadron angular correlations in $p+p$ collisions at $\sqrt{s} = 510$ GeV.” Phys. Rev. D 95, 072002 (2017).

Awards and Fellowships

RHIC/AGS Users’ Executive Committee Merit Award	05/2019
Young PHENIXian award	12/2015
Michigan Graduate 1 st Year Fellowship	08/2013-08/2014
Summa Cum Laude honors upon graduation from University of Kentucky	05/2013
Sigma Pi Sigma Physics Honor Society	05/2013
University of Kentucky Outstanding Senior Award	05/2013

Conference and Workshop Organization

1. Streaming Readout 9 Workshop, virtual, December, 2021.
2. 2021 CFNS Summer School Software Lectures, virtual, August 9-23, 2021.
3. ECCE Second Simulation Workshop, virtual, May 21, 2021.
4. ECCE Simulation Workshop, virtual, April 2, 2021.
5. LHCP Conference QCD Organizing Committee, Paris, France, June 7-12, 2021.
6. Spin and EIC workshops at the RHIC All Users Meeting, Brookhaven National Laboratory, June 2-5, 2019.
7. Scientific Secretary, American Physical Society Division of Particles and Fields Meeting, Ann Arbor, MI, August 3-7, 2015.

Collaboration and Research Community Service

ECCE Software and Computing Convenor, 2021-present

Served as convenor of the software and computing working group for the ECCE proto-collaboration.

sPHENIX Publication Policy Committee, 2021-present

Served on committee developing sPHENIX publication policies and criteria.

sPHENIX TPOT Reviewer, 12/2020

Review committee for TPC Outer Tracker for sPHENIX.

DOE Office of Nuclear Physics SBIR Proposal Reviewer, 11/2020

Review SBIR proposals related to computing in Nuclear Physics.

Nuclear Physics Day on Capitol Hill, 04/2018.

Met with staff members of Michigan Senators and Representatives to discuss funding for nuclear physics research.

PHENIX Executive Council, 01/2016-01/2017.

Served as elected member to executive council, which advises PHENIX spokesperson on scientific priorities.

Invited Conference and Workshop Presentations

1. Probing QCD at High Energy and Density with Jets INT Workshop, August 16-20, 2021.
“Recent results and future directions for jet hadronization and substructure measurements.”
2. CFNS EIC Summer School, August 9-23, 2021. “ECCE software stack and detector geometry.”
3. EICUG Summer Meeting, August 2-6, 2021. “ECCE software.”

4. RHIC Science Program Toward EIC in the Coming Years, May 24-26, 2021. “Photon-jet and dijet probes at RHIC towards the EIC.”
5. **Plenary:** Computing in High Energy Physics (CHEP) Conference, May 17-21, 2021. “Implementation of ACTS into sPHENIX Track Reconstruction.”
6. Streaming Readout VIII Workshop, April 28-30, 2021. “ORNL Streaming Readout Plans at the EIC.”
7. Resummation, Evolution, Factorization Workshop, December 7-11, 2020. “Experimental perspective on hadronization.”
8. Jets for 3D Imaging Workshop, November 23-25, 2020. “Jet substructure at the EIC.”
9. ACTS Workshop, May 25-29, 2020. “sPHENIX Experience with Acts.”
10. Second EIC Yellow Report Workshop at Pavia University, May 20-22, 2020. “Inclusive and heavy flavor jet substructure at the EIC.”
11. Connecting The Dots Workshop, April 22-24, 2020. “Requirements, status and plans for track reconstruction of the sPHENIX experiment.”
12. 3rd JETSCAPE Workshop, March 18-20, 2020. “Hadronization and jet substructure at RHIC and the LHC.”
13. 13th International Workshop on High p_T Physics in the RHIC/LHC Era, March 19-22, 2019. “Jet hadronization at LHCb.”
14. Workshop on Novel Probes of the Nucleon Structure in SIDIS, e+e- and pp (FF2019), March 14-16, 2019. “Factorization breaking, color entanglement, and hadronization of jets.”
15. Workshop on the Definition of Jets, Brookhaven National Laboratory, June 25-27, 2018. “Probing effects from QCD color with photon-jet and dijet correlations.”
16. CMS SMP-J Workshop Annual Workshop, January 25, 2017. “Color entanglement and color coherence.”

Seminars and Colloquia

1. Oak Ridge National Laboratory seminar, June 16, 2021. “Addressing Data Reduction Challenges with Next Generation Scientific Software,”
2. Workflow Workshop Seminar Series, August 14, 2020. “Data processing workflows at scattering user facilities.”
3. University of Tennessee HEP/Nuclear/Astro Seminar, February 3, 2020. “Jet substructure at RHIC and the LHC.”
4. University of Kentucky HEP/Nuclear Seminar, November 7, 2019. “Hadronization and jet substructure at RHIC and the LHC.”

5. University of Michigan HEP/Astro/Nuclear Seminar, April 8, 2019. “Jet substructure at RHIC and the LHC.”
6. Oak Ridge National Laboratory Seminar, February 18, 2019. “Peering inside protons and nuclei.”
7. Wayne State University PAN Seminar, September 14, 2018. “Effects from color flow in proton-proton and proton-nucleus collisions.”
8. Paul Laurence Dunbar High School Senior Seminar, August 31, 2018. “Career paths with a physics degree.”
9. University of Illinois HEP/MEP Colloquium, October 23, 2017. “Effects from color entanglement in proton-proton and proton-nucleus collisions.”
10. University of Michigan HEP/Astro/Nuclear Seminar, November 21, 2016. “Recent experimental results on QCD factorization breaking at RHIC.”
11. Brookhaven National Laboratory Nuclear Seminar Series, October 25, 2016. “Recent experimental results on QCD factorization breaking at RHIC.”
12. Seminar at Columbia University, October 24, 2016. “Recent experimental results on QCD factorization breaking at RHIC.”

Conference and Workshop Presentations

1. ECCE Simulation Workshop, virtual, May 21, 2021. “Developing an analysis package for ECCE DST analysis.”
2. ECCE Simulation Workshop, virtual, April 2, 2021. “Analysis in Fun4All.”
3. sPHENIX Collaboration Meeting, Boulder, Colorado (virtual), July 6, 2020. “ACTS Tracking.”
4. sPHENIX Software Mega-Workfest, Brookhaven National Laboratory, January 13-17, 2020. “Building a Fun4All Analysis Package Tutorial.”
5. sPHENIX Collaboration Meeting, Brookhaven National Laboratory, June 5-6, 2018. “Photons and clustering in sPHENIX.”
6. Quark Matter 2018, Lido, Italy, May 13-19, 2018. “PHENIX results on jet modification with π^0 - and photon-triggered two particle correlations in $p+p$, $p(d)+Au$, and $Au(Cu)+Au$ collisions.”
7. PHENIX Collaboration Meeting, Brookhaven National Laboratory, December 1-3, 2017. “High p_T correlations analysis highlights and future plans.”
8. RHIC Users Meeting Proton Structure Workshop, Brookhaven National Lab, June 20-23, 2017. “Partonic structure of nucleons and nuclei at sPHENIX.”
9. Quark Matter 2017, Chicago, Illinois, February 5-11, 2017. “Study of cold and hot nuclear matter effects on jets with direct photon-triggered correlations from PHENIX.”

10. 22nd International Spin Symposium, Urbana, IL, September 25-30, 2016. “Nonperturbative transverse momentum effects in dihadron and direct photon-hadron angular correlations.”
11. 4th Workshop on the QCD Structure of the Nucleon (QCD-N’16), Getxo, Spain, July 11-15, 2016. “Nonperturbative transverse momentum effects in dihadron and direct photon-hadron angular correlations.”
12. American Physical Society Division of Nuclear Physics Meeting, Sante Fe, NM, October 28-31, 2015. “Measuring intrinsic partonic transverse momentum via two-particle correlations in PHENIX.”
13. American Physical Society Division of Particles and Fields Meeting, Ann Arbor, MI, August 3-7, 2015. “Parton dynamics at PHENIX.”
14. Physics Graduate Student Symposium, July 8, 2015. “Partonic dynamics in high energy proton-proton collisions at PHENIX.”
15. American Physical Society Southeastern Section Meeting, Tallahassee, FL, November 14-17, 2012. “Noise analysis of the Forward GEM Tracker at STAR.”
16. P.L. Dunbar Math Science and Technology Research Symposium, Lexington, KY, April 26, 2009. “Neutrinos and shock reignition in the gain region of type IIa supernovae.”

Other Public Notes with Significant Contribution

1. sPHENIX Conceptual Design Report (CDR). The sPHENIX Collaboration.
<https://indico.bnl.gov/event/4640>, 2018.
2. Detector design study for an EIC detector around the sPHENIX solenoid. C. Aidala et al.
<https://indico.bnl.gov/event/5283/>, 2018.
3. sPHENIX medium energy barrel physics. The sPHENIX Collaboration.
<https://indico.bnl.gov/event/3866/>, 2017.
4. T1044-2017 sPHENIX test beam EMCAL analysis. J. D. Osborn and J. Huang.
<https://indico.bnl.gov/event/3854/>, 2017.
5. sPHENIX modest forward upgrade LOI. The sPHENIX Collaboration.
<https://indico.bnl.gov/event/3867/>, 2017.
6. **Nonperturbative transverse momentum effects in dihadron and direct photon-hadron angular correlations**, J. D. Osborn for the PHENIX Collaboration. Proceedings for the SPIN 2016 Symposium, Urbana, IL, September 25-30 2016. arXiv:1701.00681.
7. **Parton dynamics at PHENIX**, J. D. Osborn for the PHENIX Collaboration. Proceedings for the American Physical Society Division of Particles and Fields Conference, Ann Arbor, MI, August 4-8 2015. arXiv:1511.00016.

Student Advising and Mentorship

Graduate Students Supervised

Sebastian Araya - Track pattern recognition in ECCE.

Dillon Fitzgerald - Heavy flavor jet substructure and hadronization at the EIC.

Kara Mattioli - Heavy flavor jet tagging and substructure at LHCb.

Jordan Roth - Z^0 -hadron and Z^0 -jet correlations at LHCb.

Nicole Lewis - π^0 , η , and direct photon single spin asymmetries at PHENIX.

University Research Opportunity Program (UROP) at University of Michigan, 09/2018-05/2019

Propose and guide a research project to an undergraduate student. Supervised Brandon Liang in a jet substructure Monte Carlo study to better understand nonperturbative contributions to jets.

Additional Undergraduate Students Supervised

Nikhil Shankar, Hayden Hansen, Ezra Lesser, Nick Melekian, Ruby Araj, Emily Camras, Robert Read, Robert Cernak, Aaron White.

Teaching

University of Michigan

1. Physics for the Life Sciences Laboratory 1: Fall 2013, Spring 2014, Fall 2014, Fall 2017

University of Kentucky

1. Physics 241 - General University Physics Laboratory: Fall 2012
2. Arts and Sciences Wired Course - Measuring Science: Fall 2011

Complete Publication List

For the most up-to-date list, search ‘find a J. D. Osborn’ on inspirehep.net

Papers Submitted for Peer Reviewed Publication or on arXiv

1. ***A Common Tracking Software Project.*** X. Ai et al., arXiv:2106.13593. Submitted to Computing and Software for Big Science.

PHENIX and LHCb Collaboration Papers

(significant contribution made to those in ***bold italic text***)

1. Correlations of $\mu\mu$, $e\mu$, and ee pairs in $p+p$ collisions at $\sqrt{s} = 200$ GeV and implications for $c\bar{c}$ and $b\bar{b}$ production mechanisms. C. Aidala et al. arXiv:1805.04075. Submitted to Phys. Rev. D.
2. An Upgrade Proposal from the PHENIX Collaboration. A. Adare et al. (PHENIX Collaboration). arXiv:1501.06197.
3. Concept for an Electron Ion Collider (EIC) detector built around the BaBar solenoid. A. Adare et al. (PHENIX Collaboration). arXiv:1402.1209.

Peer Reviewed Publications

1. ***Implementation of ACTS into sPHENIX track reconstruction.*** J. D. Osborn et al. Computing and Software for Big Science 5, 23 (2021).
2. ***Science Requirements and Detector Concepts for the Electron-Ion Collider: EIC Yellow Report.*** R. Abdul Khalek et al. arXiv: 2103.05419.
3. ***Design and beam test results for the 2D projective sPHENIX electromagnetic calorimeter prototype.*** C. A. Aidala et al. IEEE Transactions on Nuclear Science, vol. 68, no. 2, pp. 173-181, Feb. 2021.
4. ***Requirements, status, and plans for track reconstruction at the sPHENIX experiment.*** J. D. Osborn for the sPHENIX Collaboration, arXiv:2007.00771. Proceedings of Connecting the Dots Workshop 2020.
5. ***Jet hadronization at LHCb.*** J. D. Osborn for the LHCb Collaboration, PoS High- p_T 2019 (2020) 013.
6. ***Nuclear gluons at RHIC in a multi-observable approach.*** I. Helenius, J. Lajoie, J. D. Osborn, P. Paakkinen, H. Paukkunen. Phys. Rev. D 100, 014004 (2019).
7. ***PHENIX results on jet modification with π^0 - and photon-triggered two particle correlations in $p+p$, $p(d)+Au$, and $Au+Au$ collisions,*** J. D. Osborn for the PHENIX Collaboration. Nuclear Physics A 982, 591-594 (2019).

8. *Study of cold and hot nuclear matter effects on jets with direct photon triggered correlations from PHENIX*, J. D. Osborn for the PHENIX Collaboration. Nuclear Physics A 967, 476-479 (2017).

PHENIX and LHCb Collaboration Papers

(significant contribution made to those in ***bold italic text***)

1. Searches for 25 rare and forbidden decays of $D^+ D_s^+$ mesons. JHEP 06, (2021) 044
2. *Probing gluon spin-momentum correlations in trasversely polarized protons through midrapidity isolated direct photons in $p^\uparrow + p$ collisions at $\sqrt{s} = 200$ GeV.* Phys. Rev. Lett. 127, 162001 (2021).
3. *Transverse single-spin asymmetries of midrapidity π^0 and η mesons in polarized $p+p$ collisions at $\sqrt{s} = 200$ GeV.* Phys. Rev. D 103, 052009 (2021).
4. Transverse momentum dependent forward neutron single spin asymmetries in transversely polarized $p+p$ collisions at $\sqrt{s} = 200$ GeV. Phys. Rev. D 103, 032007 (2021).
5. First observation of the decay $\Lambda_b^0 \rightarrow \eta_c(1S)pK^-$. Phys.Rev.D 102, 112012 (2020).
6. Measurement of the branching fractions for $B^+ \rightarrow D^{*+}D^-K^+$, $B^+ \rightarrow D^{*-}D^+K^+$, and $B^0 \rightarrow D^{*-}D^0K^+$ decays. JHEP 12, 139 (2020).
7. Measurement of the shape of the $B_s^0 \rightarrow D_s^{*-}\mu^+\nu_\mu$ differential decay rate. JHEP 12, 144 (2020).
8. Strong constraints on the $K_s^0 \rightarrow \mu^+\mu^-$ branching fraction. Phys.Rev.Lett. 125, 231801 (2020).
9. Production of π^0 , and η mesons in U+U collisions at $\sqrt{s_{NN}} = 192$ GeV. Phys. Rev. C 102, 064905 (2020).
10. Search for the doubly heavy Ξ_{bc}^0 baryon via decays to $D^0 p K^- D$. JHEP 11, 095 (2020).
11. Observation of enhanced double parton scattering in proton-lead collisions at $\sqrt{s_{NN}} = 8.16$ TeV. Phys. Rev. Lett. 125, 212001 (2020).
12. Searches for low-mass dimuon resonances. JHEP 10, 156 (2020).
13. Observation of structure in the J/ψ -pair mass spectrum. Science Bulletin 65, 23, 1983-1993 (2020).
14. Search for CP violation in $\Xi_c^+ \rightarrow pK^-\pi^+$ decays using model-independent techniques. Eur. Phys. J. C 80 (2020) 10, 986.
15. *Measurement of jet-medium interactions via direct photon-hadron correlations in Au+Au and d+Au collisions at $\sqrt{s_{NN}} = 200$ GeV.* Phys. Rev. C 102, 054910 (2020).
16. Polarization and cross section of midrapidity J/ψ production in proton-proton collisions at $\sqrt{s} = 510$ GeV. Phys.Rev.D 102, 072008 (2020)

17. Production of $b\bar{b}$ at forward rapidity in $p+p$ collisions at $\sqrt{s}=510$ GeV. Phys.Rev.D 102, 092002 (2020).
18. Study of the lineshape of the $\chi_{c1}(3872)$ state. Phys.Rev.D 102, 092005 (2020).
19. First branching fraction measurement of the suppressed decay $\Xi_c^0 \rightarrow \pi^- \Lambda_c^+$. Phys. Rev. D 102, 071101 (2020).
20. First observation of the decay $B^0 \rightarrow D^0 \bar{D}^0 K^+ \pi^-$. Phys. Rev. D 102, 051102 (2020).
21. Study of the $\psi_2(3823)$ and $\chi_{c1}(3872)$ states in $B^+ \rightarrow (J/\psi \pi^+ \pi^-) K^-$ decays. JHEP 08, 123 (2020).
22. Precision measurement of the B_c^+ meson mass. JHEP 07, 123 (2020).
23. Measurement of the $\Lambda_b^0 \rightarrow J/\psi \Lambda$ angular distribution and the Λ_b^0 polarisation in pp collisions. JHEP 06, 110 (2020).
24. Observation of new Ξ_c^0 baryons decaying to $\Lambda_c^+ K^-$. Phys. Rev. Lett. 124, 222001 (2020).
25. Measurement of CP averaged observables in the $B^0 \rightarrow K^{*0} \mu^+ \mu^-$ decay. Phys. Rev. Lett. 125, 011802 (2020).
26. Search for the lepton flavour violating decay $B^+ \rightarrow K^+ \mu^- \tau^+$ using B_{s2}^{*0} decays. JHEP 06, 129 (2020).
27. Search for the rare decays $B_s^0 \rightarrow e^+ e^-$ and $B^0 \rightarrow e^+ e^-$. Phys. Rev. Lett. 124, 211802 (2020).
28. Measurement of charged pion double spin asymmetries at midrapidity in longitudinally polarized $p+p$ collisions at $\sqrt{s} = 510$ GeV. Phys. Rev. D 102, 032001 (2020).
29. Measurement of CP observables in $B^\pm \rightarrow DK^\pm$ and $B^\pm \rightarrow D\pi^\pm$ with $D \rightarrow K_s^0 K^\pm \pi^\mp$ decays. JHEP 06, 058 (2020).
30. Observation of a new baryon state in the $\Lambda_b^0 \pi^+ \pi^-$ mass spectrum. JHEP 06, 136 (2020).
31. Search for CP violation and observation of P violation in $\Lambda_b^0 \rightarrow p \pi^- \pi^+ \pi^-$ decays. Phys. Rev. D 102, 051101 (2020).
32. Measurement of the branching fraction of the decay $B_s^0 \rightarrow K_s^0 K_s^0$. Phys. Rev. D 102, 012011 (2020).
33. Measurement of J/ψ at forward and backward rapidity in $p+p$, $p+\text{Al}$, $p+\text{Au}$, and $^3\text{He}+\text{Au}$ collisions at $\sqrt{s_{NN}}=200$ GeV. Phys. Rev. C 102, 014902 (2020).
34. Measurement of $|V_{cb}|$ with $B_s^0 \rightarrow D_s^{(*)-} \mu^+ \nu_\mu$ decays. Phys. Rev. D 101, 072004 (2020).
35. Test of lepton universality with $\Lambda_b^0 \rightarrow p K^- \ell^+ \ell^-$ decays. JHEP 20, 040 (2020).
36. Measurement of CP violation in $B^0 \rightarrow D^{*\pm} D^\mp$ decays. JHEP 03, 147 (2020).
37. Observation of the semileptonic decay $B^+ \rightarrow p \bar{p} \mu^+ \nu_\mu$. JHEP 03, 146 (2020).

38. Measurement of f_s/f_u variation with proton-proton collision energy and kinematics. Phys. Rev. Lett. 124, 122002 (2020).
39. First observation of excited Ω_b^- states. Phys. Rev. Lett. 124, 082002 (2020).
40. J/ψ and $\psi(2S)$ production at forward rapidity in $p+p$ collisions at $\sqrt{s} = 510$ GeV. Phys. Rev. D 101, 052006 (2020).
41. Isospin amplitudes in $\Lambda_b^0 \rightarrow J/\psi \Lambda(\Sigma^0)$ and $\Xi_b^0 \rightarrow J/\psi \Xi^0(\Lambda)$ decays. Phys. Rev. Lett. 124, 111802 (2020).
42. Precision measurement of the Ξ_{cc}^{++} mass. JHEP 2002, 049 (2020).
43. Determination of quantum numbers for several excited charmed mesons observed in $B^- \rightarrow D^{*+} \pi^- \pi^-$ decays. Phys. Rev. D 101, 032005 (2020).
44. Measurement of the $\eta_c(1S)$ production cross-section in pp collisions at $\sqrt{s} = 13$ TeV. Eur. Phys. J. C 80, 191 (2020).
45. Nuclear modification factor of charged hadrons at forward and backward rapidity in $p+\text{Al}$ and $p+\text{Au}$ collisions at $\sqrt{s_{NN}} = 200$ GeV. Phys. Rev. C 101, 034910 (2020).
46. Updated measurement of decay-time-dependent CP asymmetries in $D^0 \rightarrow K^+ K^-$ and $D^0 \rightarrow \pi^+ \pi^-$ decays. R. Aaij et al. Phys.Rev. D101, 012005 (2020).
47. Measurement of Ξ_{cc}^{++} production in pp collisions at $\sqrt{s} = 13$ TeV. Chin. Phys. C 44, 022001 (2020).
48. Amplitude analysis of the $B^+ \rightarrow \pi^+ \pi^+ \pi^-$ decay. R. Aaij et al. Phys.Rev. D 101, 012006 (2020).
49. Observation of several sources of CP violation in $B^+ \rightarrow \pi^+ \pi^+ \pi^-$ decays. R. Aaij et al. Phys.Rev.Lett. 124, 031801 (2020).
50. Search for $A' \rightarrow \mu^+ \mu^-$ decays. R. Aaij et al. Phys. Rev. Lett. 124, 041801 (2020).
51. Search for the doubly charmed baryon Ξ_{cc}^+ . Sci. China Phys. Mech. Astron. 63, 221062 (2020).
52. Measurement of the electron reconstruction efficiency at LHCb. JINST 14, P11023 (2019).
53. Measurement of the B_c^- meson production fraction asymmetry in 7 and 13 TeV pp collisions. R. Aaij et al. Phys.Rev. D 100, 112006 (2019).
54. Search for the lepton-flavour violating decays $B^+ \rightarrow K^+ \mu^\pm e^\mp$. R. Aaij et al. Phys.Rev.Lett. 123, 241802 (2019).
55. Measurement of CP violation in the $B_s^0 \rightarrow \phi\phi$ decay and search for the $B^0 \rightarrow \phi\phi$ decay. R. Aaij et al. JHEP 1909, 028 (2019).
56. **Measurement of charged-hadron production in Z-tagged jets in proton-proton collisions at $\sqrt{s} = 8$ TeV.** Phys. Rev. Lett. 123, 232001 (2019).
57. Observation of new resonances in the $\Lambda_b^0 \pi^+ \pi^-$ system. Phys. Rev. Lett. 123, 152001 (2019)

58. Search for the lepton-flavour-violating decays $B_s^0 \rightarrow \tau^\pm \mu^\mp$ and $B^0 \rightarrow \tau^\pm \mu^\mp$. Phys. Rev. Lett. 123, 211801 (2019).
59. A search for $\Xi_{cc}^{++} \rightarrow D^+ p K^- \pi^+$ decays. JHEP 1910, 124 (2019).
60. Observation of the $\Lambda_b^0 \rightarrow \chi_{c1}(3872) p K^-$ decay. JHEP 1909, 028 (2019).
61. Measurement of CP observables in the process $B^0 \rightarrow DK^{*0}$ with two- and four-body D decays. JHEP 1909, 041 (2019).
62. Updated measurement of time-dependent CP -violating observables in $B_s^0 \rightarrow J/\psi K^+ K^-$ decays. Eur. Phys. J. C 79, 706 (2019).
63. Precision measurement of the Λ_c^+ , Ξ_c^+ , and Ξ_c^0 baryon lifetimes. Phys. Rev. D 100, 032001 (2019).
64. Measurement of CP -violating and mixing-induced observables in $B_s^0 \rightarrow \phi \gamma$ decays. Phys. Rev. Lett. 123, 081802 (2019).
65. First observation of the radiative decay $\Lambda_b^0 \rightarrow \Lambda \gamma$. Phys. Rev. Lett. 123, 031801 (2019).
66. Nuclear dependence of the transverse single-spin asymmetry in the production of charged hadrons at forward rapidity in polarized $p + p$, $p + \text{Al}$, and $p + \text{Au}$ collisions at $\sqrt{s_{NN}} = 200$ GeV. Phys. Rev. Lett. 123, 122001 (2019).
67. Amplitude analysis of the $B_{(s)}^0 \rightarrow K^{*0} \bar{K}^{*0}$ decay and measurement of its relative branching fraction relative to the $B_s^0 \rightarrow K^{*0} \bar{K}^{*0}$. JHEP 1907, 032 (2019).
68. Near-threshold $D\bar{D}$ spectroscopy and observation of a new charmonium state. JHEP 1907, 035 (2019).
69. Measurement of the mass difference between neutral charm-meson eigenstates. Phys. Rev. Lett. 122, 231802 (2019).
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